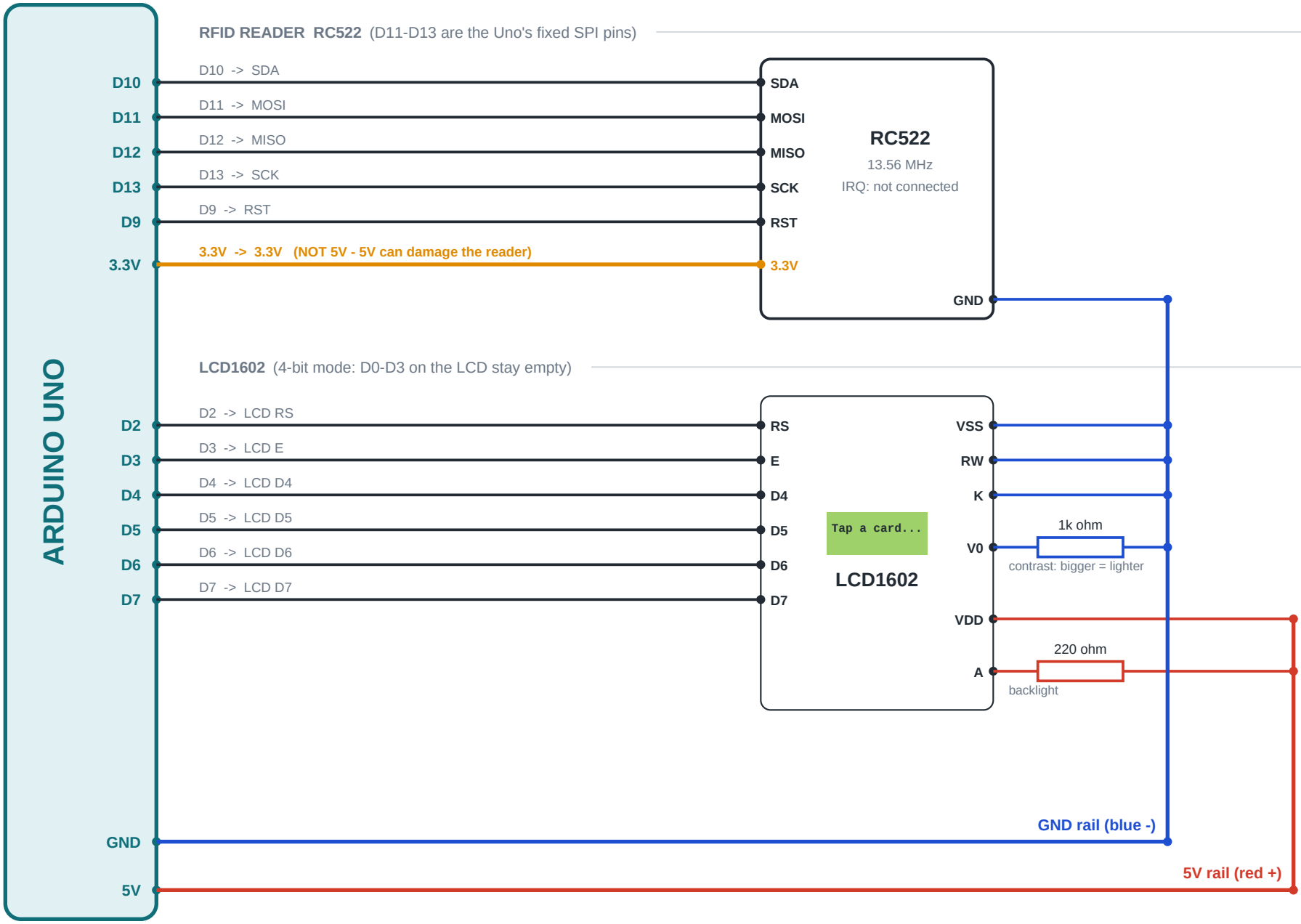
RFID Music Box - Phase 1 Wiring

Arduino Uno | RC522 RFID reader + LCD1602 | Sketch: src/main.cpp



Lines that cross without a dot are not connected. The RC522's 3.3V is the only wire that does not use a rail.

RFID Music Box - Phase 1 Parts & Connections

Everything on page 1 as lists, plus a few tips.

Parts

Qty	Part	Notes
1	Arduino Uno + USB cable	Powers everything.
1	RC522 RFID module, card, key fob	From the kit. Solder the header pins if they are loose.
1	LCD1602 module	16 pins along the top edge.
1	1k ohm resistor	Brown, black, red. LCD contrast.
1	220 ohm resistor	Red, red, brown. LCD backlight.
1	Breadboard	Uno 5V to the red (+) rail, GND to the blue (-) rail.
~22	Jumper wires	Female-to-male for the RC522 if it has header pins.

Connections

Uno pin	Goes to	Notes
D10 / D9	RC522 SDA / RST	
D11 / D12 / D13	RC522 MOSI / MISO / SCK	Hardware SPI.
3.3V	RC522 3.3V	Not 5V. Power header, between RESET and 5V.
D2 / D3	LCD RS / E	
D4 - D7	LCD D4 - D7	Same number on both ends.
GND rail	RC522 GND, LCD VSS, RW, K	RW to GND = write-only.
GND rail	1k ohm, then LCD V0	Contrast.
5V rail	LCD VDD; 220 ohm, then LCD A	Power and backlight.

If something is wrong

Backlight on, no text: contrast. Faint or nothing: use a smaller resistor (470 or 220 ohm), or wire V0 straight to GND. Solid blocks on line 1: use a bigger one (2.2k).

Solid blocks that never change: the sketch isn't talking to the LCD. Check RS, E and D4-D7, and that the upload worked.

Nothing when you tap a card: check the reader gets 3.3V, the SPI wires are on D10-D13, and the header is soldered. Hold the card flat, within 2 cm.

LCD says RFID not found: the Uno asked the reader for its version and got no answer. Check all 7 RC522 wires, especially 3.3V and GND, then press the Uno's reset button.

Serial Monitor shows gibberish: set it to 9600 baud (pio device monitor already does).

Card shows once, then not again: by design. Lift it off the reader and tap again.

ID changes on every tap: phones and some bank cards use random IDs. Use the kit's card, fob or MIFARE/NTAG tags.

What it does

While idle the LCD shows **Tap a card...** On a tap the LCD shows **Card:** and the ID for 3 seconds, and the Serial Monitor prints **Card UID: A3 4F 12 9C**. Write each ID down: phase 2 maps them to music folders.